

# E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai

Accredited by NAAC with 'A++' Grade

Accredited by NBA (CSE, EEE, MECH, IT, CIVIL, ECE)

NAGAPATTINAM – 611002



## B.TECH. INFORMATION TECHNOLOGY

**R-2019**

### Full Time Curriculum and Syllabus

Total Credits to be earned for the award of the Degree: **160 (Regular); 124 (Lateral Entry)**

Batch: 2019-2023 (Admitted from 2019 onwards)

### FIRST YEAR

| SEMESTER I       |                                                          |    |   |   |    |               |     |       |          |
|------------------|----------------------------------------------------------|----|---|---|----|---------------|-----|-------|----------|
| Course Code      | Course Name                                              | L  | T | P | C  | Maximum Marks |     |       | Category |
|                  |                                                          |    |   |   |    | CA            | ES  | Total |          |
| Theory Course    |                                                          |    |   |   |    |               |     |       |          |
| 1901MA102        | Engineering Mathematics –I (Calculus and Linear Algebra) | 3  | 2 | 0 | 4  | 40            | 60  | 100   | BSC      |
| 1901CH101        | Applied Chemistry in Informatics                         | 3  | 0 | 0 | 3  | 40            | 60  | 100   | BSC      |
| 1901GEX03        | Programming for Problem Solving                          | 3  | 0 | 0 | 3  | 40            | 60  | 100   | ESC      |
| 1901ENX01        | English for Engineers                                    | 2  | 0 | 0 | 2  | 40            | 60  | 100   | HSSC     |
| Practical Course |                                                          |    |   |   |    |               |     |       |          |
| 1901GEX52        | Computer Programming Lab                                 | 0  | 0 | 2 | 1  | 50            | 50  | 100   | ESC      |
| 1901GE151        | Engineering Intelligence I                               | 0  | 0 | 2 | 1  | 100           | 0   | 100   | EEC      |
| 1901CHX51        | Engineering Chemistry Lab                                | 0  | 0 | 2 | 1  | 50            | 50  | 100   | BSC      |
| 1901HSX51        | Communication Skill Lab                                  | 0  | 0 | 2 | 1  | 50            | 50  | 100   | HSSC     |
| Total            |                                                          | 11 | 2 | 8 | 16 | 410           | 390 | 800   |          |

| SEMESTER II      |                                              |    |   |    |    |               |     |       |          |
|------------------|----------------------------------------------|----|---|----|----|---------------|-----|-------|----------|
| Course Code      | Course Name                                  | L  | T | P  | C  | Maximum Marks |     |       | Category |
|                  |                                              |    |   |    |    | CA            | ES  | Total |          |
| Theory Course    |                                              |    |   |    |    |               |     |       |          |
| 1901MA202        | Engineering Mathematics –II                  | 3  | 2 | 0  | 4  | 40            | 60  | 100   | BSC      |
| 1901PH201        | Physics for Information Science              | 3  | 0 | 0  | 3  | 40            | 60  | 100   | BSC      |
| 1901GEX01        | Basic Electrical and Electronics Engineering | 3  | 0 | 0  | 3  | 40            | 60  | 100   | ESC      |
| 1901GEX02        | Engineering Graphics                         | 2  | 0 | 2  | 3  | 50            | 50  | 100   | ESC      |
| 1901GE201        | Engineering Exploration                      | 2  | 0 | 0  | 2  | 100           | 0   | 100   | ESC      |
| Practical Course |                                              |    |   |    |    |               |     |       |          |
| 1901GE254        | Computer Hardware and IT Essentials Lab      | 0  | 0 | 2  | 1  | 50            | 50  | 100   | ESC      |
| 1901GE252        | Engineering Intelligence - II                | 0  | 0 | 2  | 1  | 100           | 0   | 100   | EEC      |
| 1901GEX51        | CAD Lab                                      | 0  | 0 | 2  | 1  | 50            | 50  | 100   | ESC      |
| 1901GEX53        | Basic Electrical and Electronics Engineering | 0  | 0 | 2  | 1  | 50            | 50  | 100   | ESC      |
| 1901PHX51        | Engineering Physics Lab                      | 0  | 0 | 2  | 1  | 50            | 50  | 100   | BSC      |
| Total            |                                              | 13 | 2 | 12 | 20 | 570           | 430 | 1000  |          |

## SECOND YEAR

| SEMESTER III      |                                                                |    |   |   |    |               |     |       |          |
|-------------------|----------------------------------------------------------------|----|---|---|----|---------------|-----|-------|----------|
| Course Code       | Course Name                                                    | L  | T | P | C  | Maximum Marks |     |       | Category |
|                   |                                                                |    |   |   |    | CA            | ES  | Total |          |
| Theory Course     |                                                                |    |   |   |    |               |     |       |          |
| 1901MA302         | Engineering Mathematics – III(Queuing Model and Network Model) | 3  | 2 | 0 | 4  | 40            | 60  | 100   | BS       |
| 1902IT301         | Data Structures and Algorithms                                 | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |
| 1902IT302         | Computer Organization and Architecture                         | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |
| 1902IT303         | Digital Principles and Design                                  | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |
| 1902IT304         | Problem solving using Python                                   | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |
| 1901GEX04         | Biology for Engineers                                          | 3  | 0 | 0 | 3  | 40            | 60  | 100   | BS       |
| Laboratory Course |                                                                |    |   |   |    |               |     |       |          |
| 1902IT351         | Data Structures and Algorithms Lab                             | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |
| 1902IT352         | Digital Principles and Design Lab                              | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |
| 1902IT353         | Python Programming Lab                                         | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |
| 1904GE351         | Life Skills: Soft Skill                                        | 0  | 0 | 2 | 1  | 100           | -   | 100   | EEC      |
| Audit Course      |                                                                |    |   |   |    |               |     |       |          |
| 1902MCX02         | Constitution of India                                          | 2  | 0 | 0 | 0  | 100           | -   | 100   | MC       |
| Total             |                                                                | 20 | 2 | 8 | 23 | 590           | 510 | 1100  | -        |

| SEMESTER IV       |                                             |    |   |   |    |               |     |       |          |  |
|-------------------|---------------------------------------------|----|---|---|----|---------------|-----|-------|----------|--|
| Course Code       | Course Name                                 | L  | T | P | C  | Maximum Marks |     |       | Category |  |
|                   |                                             |    |   |   |    | CA            | ES  | Total |          |  |
| Theory Course     |                                             |    |   |   |    |               |     |       |          |  |
| 1902IT401         | Database Management Systems                 | 3  | 0 | 0 | 3  | 40            | 60  | 100   | BS       |  |
| 1902IT402         | Java Programming                            | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
| 1902IT403         | Operating Systems                           | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
| 1902IT404         | Software Engineering and Project Management | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
| 1902IT405         | Computer Networks                           | 3  | 2 | 0 | 4  | 40            | 60  | 100   | PC       |  |
| 1902IT406         | Principles of Communication                 | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
| Laboratory Course |                                             |    |   |   |    |               |     |       |          |  |
| 1902IT451         | Database Management Systems Lab             | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |  |
| 1902IT452         | Java Programming Lab                        | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |  |
| 1902IT453         | Operating Systems Lab                       | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |  |
| 1904GE451         | Life Skills: Verbal Ability                 | 0  | 0 | 2 | 1  | 100           | -   | 100   | EEC      |  |
| Audit Course      |                                             |    |   |   |    |               |     |       |          |  |
| 1901MCX01         | Environmental Science                       | 2  | 0 | 0 | 0  | 100           | -   | -     | MC       |  |
| Total             |                                             | 20 | 2 | 8 | 23 | 590           | 510 | 1100  | -        |  |

## THIRD YEAR

| SEMESTER V    |                                     |   |   |   |   |               |     |       |          |
|---------------|-------------------------------------|---|---|---|---|---------------|-----|-------|----------|
| Course Code   | Course Name                         | L | T | P | C | Maximum Marks |     |       | Category |
|               |                                     |   |   |   |   | CA            | E S | Total |          |
| Theory Course |                                     |   |   |   |   |               |     |       |          |
| 1902IT501     | Object Oriented Analysis and Design | 3 | 0 | 0 | 3 | 40            | 60  | 100   | PC       |
| 1902IT502     | Web Programming                     | 3 | 0 | 0 | 3 | 40            | 60  | 100   | PC       |
| 1902IT503     | Security in Computing               | 3 | 2 | 0 | 4 | 40            | 60  | 100   | PC       |
| 1902IT504     | Internet of Things                  | 3 | 0 | 0 | 3 | 40            | 60  | 100   | PC       |
|               | Professional Core Elective I        | 3 | 0 | 0 | 3 | 40            | 60  | 100   | PE       |
|               | Humanities Science Elective I       | 3 | 0 | 0 | 3 | 40            | 60  | 100   | HSSE     |

| <b>Laboratory Course</b> |                                         |    |   |   |    |     |     |      |     |  |
|--------------------------|-----------------------------------------|----|---|---|----|-----|-----|------|-----|--|
| 1902IT551                | Web Programming Lab                     | 0  | 0 | 2 | 1  | 50  | 50  | 100  | PC  |  |
| 1904IT551                | Case Tools (Mini Project I)             | 0  | 0 | 2 | 1  | 50  | 50  | 100  | EEC |  |
| 1904GE551                | Life Skills: Aptitude I                 | 0  | 0 | 2 | 1  | 100 | -   | 100  | EEC |  |
| 1904IT552                | Startup Opportunities for IT Engineers  | 0  | 0 | 2 | 1  | 100 | -   | 100  | EEC |  |
| <b>Audit Course</b>      |                                         |    |   |   |    |     |     |      |     |  |
| 1901MCX03                | Essence of Indian Traditional Knowledge | 2  | 0 | 0 | -  | 100 | -   | 100  | AC  |  |
| Total                    |                                         | 20 | 2 | 8 | 23 | 640 | 460 | 1100 | -   |  |

| SEMESTER VI       |                                                  |    |   |   |    |               |     |       |          |  |
|-------------------|--------------------------------------------------|----|---|---|----|---------------|-----|-------|----------|--|
| Course Code       | Course Name                                      | L  | T | P | C  | Maximum Marks |     |       | Category |  |
|                   |                                                  |    |   |   |    | CA            | ES  | Total |          |  |
| Theory Course     |                                                  |    |   |   |    |               |     |       |          |  |
| 1902IT601         | Artificial Intelligence                          | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
| 1902IT602         | Mobile Computing                                 | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
| 1902IT603         | Data Warehousing and Data Mining                 | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PC       |  |
|                   | Professional Core Elective II                    | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PE       |  |
|                   | Humanities Science Elective II                   | 3  | 0 | 0 | 3  | 40            | 60  | 100   | HSSE     |  |
|                   | Open Elective- I                                 | 3  | 0 | 0 | 3  | 40            | 60  | 100   | OE       |  |
| Laboratory Course |                                                  |    |   |   |    |               |     |       |          |  |
| 1902IT651         | Data Mining Lab                                  | 0  | 0 | 2 | 1  | 50            | 50  | 100   | PC       |  |
| 1904IT651         | Mobile Application Development (Mini Project II) | 0  | 0 | 2 | 1  | 50            | 50  | 100   | EEC      |  |
| 1904GE651         | Life Skills: Aptitude II                         | 0  | 0 | 2 | 1  | 100           | -   | 100   | EEC      |  |
| 1904IT652         | Industrial Visit Presentation                    | 0  | 0 | 0 | 1  | 100           | -   | 100   | EEC      |  |
| Total             |                                                  | 18 | 0 | 6 | 22 | 540           | 460 | 1000  |          |  |

## **FINAL YEAR**

| SEMESTER VII      |                                         |    |   |   |    |               |     |       |          |  |
|-------------------|-----------------------------------------|----|---|---|----|---------------|-----|-------|----------|--|
| Course Code       | Course Name                             | L  | T | P | C  | Maximum Marks |     |       | Category |  |
|                   |                                         |    |   |   |    | CA            | ES  | Total |          |  |
| Theory Course     |                                         |    |   |   |    |               |     |       |          |  |
| 1901MGX07         | Universal Human Values and Ethics       | 3  | 0 | 0 | 3  | 40            | 60  | 100   | HSSC     |  |
| 1902IT701         | Cloud Computing                         | 3  | 2 | 0 | 4  | 50            | 50  | 100   | PC       |  |
|                   | Professional Core Elective III          | 3  | 0 | 0 | 3  | 40            | 60  | 100   | PE       |  |
|                   | Humanities Science Elective III         | 3  | 0 | 0 | 3  | 40            | 60  | 100   | HSSE     |  |
|                   | Open Elective - II                      | 3  | 0 | 0 | 3  | 40            | 60  | 100   | OE       |  |
| Laboratory Course |                                         |    |   |   |    |               |     |       |          |  |
| 1904IT751         | Software Development (Mini Project III) | 0  | 0 | 2 | 1  | 50            | 50  | 100   | EEC      |  |
| 1904GE751         | Life Skills: Comprehensive              | 2  | 0 | 0 | 2  | 100           | -   | 100   | EEC      |  |
| 1904IT752         | Inplant Training/Internship             | 0  | 0 | 0 | 1  | 100           | -   | 100   | EEC      |  |
| Total             |                                         | 17 | 2 | 2 | 20 | 460           | 360 | 800   | -        |  |

| SEMESTER VIII     |                               |   |   |    |    |               |     |       |          |  |
|-------------------|-------------------------------|---|---|----|----|---------------|-----|-------|----------|--|
| Course Code       | Course Name                   | L | T | P  | C  | Maximum Marks |     |       | Category |  |
|                   |                               |   |   |    |    | CA            | ES  | Total |          |  |
| Theory Course     |                               |   |   |    |    |               |     |       |          |  |
|                   | Professional Core Elective IV | 3 | 0 | 0  | 3  | 40            | 60  | 100   | PE       |  |
|                   | Professional Core Elective V  | 3 | 0 | 0  | 3  | 40            | 60  | 100   | PE       |  |
| Laboratory Course |                               |   |   |    |    |               |     |       |          |  |
| 1904IT851         | Project Work                  | 0 | 0 | 14 | 7  | 50            | 50  | 100   | EEC      |  |
| Total             |                               | 6 | 0 | 14 | 13 | 130           | 170 | 300   | -        |  |

## List of Electives for / offered by B.TECH – INFORMATION TECHNOLOGY

### Professional Elective Courses (PEC)

| Course Code                         | Course Name                        | L | T | P | C | Maximum Marks |    |       | Category |
|-------------------------------------|------------------------------------|---|---|---|---|---------------|----|-------|----------|
|                                     |                                    |   |   |   |   | CA            | ES | Total |          |
| Professional Elective Courses - I   |                                    |   |   |   |   |               |    |       |          |
| 1903IT001                           | Software Testing Methods and Tools | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT002                           | Virtual Reality                    | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT003                           | Information Theory and Coding      | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT004                           | Mainframe Computing                | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| Professional Elective Courses - II  |                                    |   |   |   |   |               |    |       |          |
| 1903IT005                           | Multicore Architecture             | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT006                           | Cyber Security                     | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT007                           | Compiler Design                    | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT008                           | Wireless Communication             | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| Professional Elective Courses - III |                                    |   |   |   |   |               |    |       |          |
| 1903IT009                           | Robot Programming                  | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT010                           | Computer Vision                    | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT011                           | Information Management             | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT012                           | Software Architectures             | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| Professional Elective Courses - IV  |                                    |   |   |   |   |               |    |       |          |
| 1903IT013                           | Enterprise Resource Planning       | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT014                           | Business Intelligence              | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT015                           | Data Analytics                     | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT016                           | Social Network Analysis            | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| Professional Elective Courses - V   |                                    |   |   |   |   |               |    |       |          |
| 1903IT017                           | Human Computer Interaction         | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT018                           | Blockchain Technologies            | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT019                           | Game Programming                   | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |
| 1903IT020                           | Deep Learning                      | 3 | 0 | 0 | 3 | 40            | 60 | 100   | PEC      |

### HUMANITIES, SOCIAL SCIENCE AND MANAGEMENT ELECTIVE COURSES (HSSEC) (9)

| S. No.     | Course Code | Name of the Course                           | Number of Hours / Week |   |   | Credit |
|------------|-------------|----------------------------------------------|------------------------|---|---|--------|
|            |             |                                              | L                      | T | P |        |
| Elective 1 |             |                                              |                        |   |   |        |
| 1.         | 1901MGX01   | Total Quality Management                     | 3                      | 0 | 0 | 3      |
| 2.         | 1901MGX04   | Principles of Management                     | 3                      | 0 | 0 | 3      |
| 3.         | 1901MGX05   | Engineering Economics and Finance            | 3                      | 0 | 0 | 3      |
| 4.         | 1901MGX06   | Human Resource Development in IT             | 3                      | 0 | 0 | 3      |
| Elective 2 |             |                                              |                        |   |   |        |
| 1.         | 1901HS001   | Innovation and Entrepreneurship Fundamentals | 3                      | 0 | 0 | 3      |
| 2.         | 1901HS002   | Intellectual Property Rights for Engineers   | 3                      | 0 | 0 | 3      |
| 3.         | 1901HS003   | Startup Entrepreneurship                     | 3                      | 0 | 0 | 3      |
| 4.         | 1901HS004   | Business Model Innovation                    | 3                      | 0 | 0 | 3      |

| Elective 3 |           |                                       |   |   |   |   |
|------------|-----------|---------------------------------------|---|---|---|---|
| 1.         | 1901HS005 | Social Entrepreneurship               | 3 | 0 | 0 | 3 |
| 2.         | 1901HS006 | Design Thinking for Innovation        | 3 | 0 | 0 | 3 |
| 3.         | 1901HS007 | Indian Patent Law and Patent Drafting | 3 | 0 | 0 | 3 |
| 4.         | 1901HS008 | IP Management and Commercialization   | 3 | 0 | 0 | 3 |

#### ONE CREDIT COURSES (OCC)

| Course    | Course Name                                   | Category |
|-----------|-----------------------------------------------|----------|
| 1905IT001 | Full Stack Developer                          | EEC      |
| 1905IT002 | Devops                                        | EEC      |
| 1905IT003 | Angular JS                                    | EEC      |
| 1905IT004 | Sales force Trail Head                        | EEC      |
| 1905IT005 | VMware Data Centre and Virtualization         | EEC      |
| 1905IT006 | R-Programming                                 | EEC      |
| 1905IT007 | EMC Cloud Infrastructure and Services         | EEC      |
| 1905IT008 | Amazon Web Services Cloud Computing Architect | EEC      |
| 1905IT009 | Agriculture for Engineers                     | EEC      |
| 1905IT010 | Mongo and Maria Db                            | EEC      |
| 1905IT011 | Building Smart Cities                         | EEC      |
| 1905IT012 | Industrial Safety for Engineers               | EEC      |
| 1905IT013 | Introduction to Fisheries Engineering         | EEC      |
| 1905IT014 | Political Science                             | EEC      |
| 1905IT015 | Visual Communication                          | EEC      |